

Steps for Sentiment Analysis Using Multi-Head Attention

Notation:

- L : Length of the input sentence.
- d : Embedding dimension.
- h : Number of attention heads.
- d_{hidden} : Hidden dimension in the feedforward network.
- C : Number of sentiment categories.

1. Input Embedding and Positional Encoding

- Convert each word in the sentence into a vector of size d , forming the matrix $X \in \mathbb{R}^{L \times d}$.
- Add positional encodings PE to obtain:

$$X' = X + PE,$$

where $PE \in \mathbb{R}^{L \times d}$. The matrix X' has dimensions (L, d) .

2. Multi-Head Attention

For each attention head j , compute:

$$Q_j = X'W_Q^{(j)}, \quad K_j = X'W_K^{(j)}, \quad V_j = X'W_V^{(j)},$$

where $W_Q^{(j)}, W_K^{(j)}, W_V^{(j)} \in \mathbb{R}^{d \times (d/h)}$. The resulting matrices Q_j, K_j, V_j have dimensions $(L, d/h)$.

Attention Score Calculation

$$Z^{(j)} = \text{softmax} \left(\frac{Q_j K_j^T}{\sqrt{d/h}} \right) V_j,$$

where $Z^{(j)} \in \mathbb{R}^{L \times (d/h)}$. Concatenate the outputs from all heads to obtain:

$$Z = [Z^{(1)}, Z^{(2)}, \dots, Z^{(h)}],$$

where $Z \in \mathbb{R}^{L \times d}$.

3. Add Residual Connection and Apply Layer Normalization

$$Z' = \text{LayerNorm}(Z + X'),$$

where $Z' \in \mathbb{R}^{L \times d}$.

4. Feedforward Neural Network

- **First Linear Layer with ReLU Activation:**

$$F(Z') = \text{ReLU}(Z'W_1 + b_1),$$

where $W_1 \in \mathbb{R}^{d \times d_{\text{hidden}}}$, $b_1 \in \mathbb{R}^{d_{\text{hidden}}}$, and $F(Z') \in \mathbb{R}^{L \times d_{\text{hidden}}}$.

- **Second Linear Layer:**

$$F(Z') = F(Z')W_2 + b_2,$$

where $W_2 \in \mathbb{R}^{d_{\text{hidden}} \times d}$, $b_2 \in \mathbb{R}^d$. Now $F(Z') \in \mathbb{R}^{L \times d}$.

- **Add Residual Connection and Apply Layer Normalization:**

$$Z_{\text{final}} = \text{LayerNorm}(F(Z') + Z').$$

5. Pooling Operation

Aggregate word representations using mean pooling:

$$\mathbf{c} = \frac{1}{L} \sum_{i=1}^L Z_{\text{final},i}.$$

The context vector $\mathbf{c}$ has dimensions (d) .

6. Final Classification Layer

Pass $\mathbf{c}$ through a fully connected layer with softmax:

$$\hat{y} = \text{softmax}(\mathbf{W}_{\text{out}}\mathbf{c} + b_{\text{out}}),$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times C}$ and $b_{\text{out}} \in \mathbb{R}^C$. The predicted vector $\hat{y}$ has dimensions (C) .

7. Loss Function (Cross-Entropy)

The cross-entropy loss function for a one-hot encoded label y and predicted probabilities $\hat{y}$ is:

$$\mathcal{L} = - \sum_{k=1}^C y_k \log(\hat{y}_k),$$

where y_k is the true label for category k and $\hat{y}_k$ is the predicted probability for category k .